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Q167: (a) $M_o = I_o \alpha$, $\therefore I_o = I_G + mr^2$ (parallel)

$$P\left(\frac{L}{2} + r\right) = \left(\frac{1}{2}mL^2 + mr^2\right)\alpha$$

$$\therefore \alpha = 107.1 \text{ rad/s}^2$$

(b) $\therefore \underline{a = \alpha r}$

$$\begin{cases} \vec{a}_t = \alpha r \\ \vec{a}_n = \omega^2 r \end{cases}$$

$$\sum F_{xG} = ma_{xG}$$

$$N_x - P = m(\alpha r)$$

$$\therefore N_x = -21.4 \text{ N}$$

$$N_y - mg = ma_n = 0$$

$$\therefore N_y = mg = 39.2 \text{ N}$$

Q16.12-15

(Pure Rolling) (a) $\epsilon = \alpha$
(b) $\mu_s m g$

$$P r_G = I_G \alpha$$

$$P r_2 = (mr_G^2 + mr_2^2)\alpha \quad \text{--- (1)}$$

$$\therefore \alpha = \frac{a_G}{r_2}, \quad a_G = a_2$$

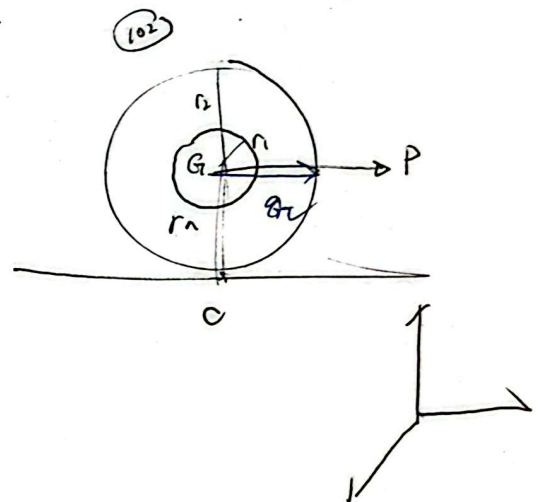
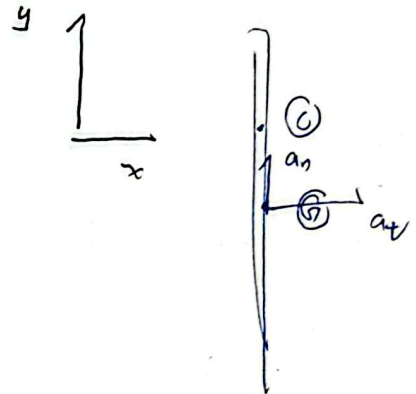
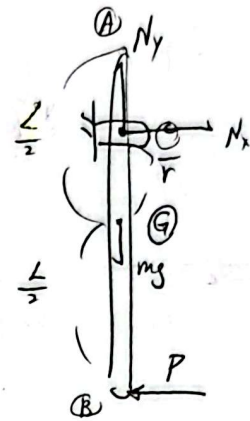
$$\therefore P - f = ma_G$$

$$f \leq \mu_s mg$$

$$\textcircled{1} \quad \alpha = 17.78 \text{ rad/s}^2 \quad \curvearrowright$$

$$\therefore a_G = \alpha r_2 = 2.13 \text{ m/s}^2, \quad f_s = 7.2 \rightarrow$$

$$\therefore \mu_s \leq 0.122, \quad \mu_{s \text{ min}} = 0.122$$



Q16.103-108

$$M_c = \bar{I}_c \alpha$$

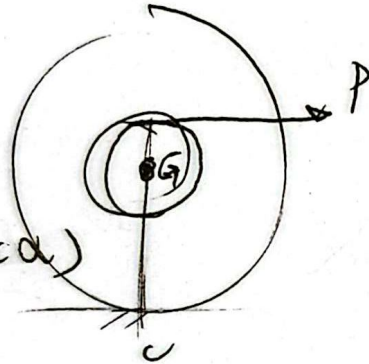
$$P(r_2 + r_1) = (m\ddot{r}_G + m\ddot{r}_1) r_1 \alpha$$

$$\alpha = 26.67 \text{ rad/s}^2$$

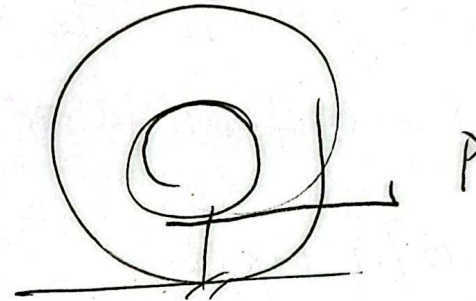
~~$$a_G = \alpha(r_2 + r_1)$$~~

$$a_G = \alpha(r_2) = 3.2 \text{ m/s}^2$$

$$\mu_{\text{min}} = 0.014$$



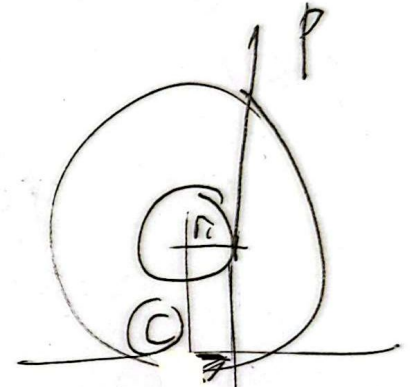
Similarly for
Q16.104, Q16.105



$$P(r_1 - r_2) = (m\ddot{r}_G + m\ddot{r}_2) r_2 \alpha$$

$$a_G = \alpha r_2$$

$$\mu_{\text{min}} = 0.23$$



$$P(r_1) = (m\ddot{r}_G + m\ddot{r}_2) r_2 \alpha$$

$$a_G = \alpha r_2$$

~~$$f \leq \mu(mg - P)$$~~

$$\therefore \mu_{\text{min}} = 0.165$$